

UNIVERSITAT POLITÈCNICA DE CATALUNYA

School of Professional & Executive Development

Postgrado en

Mobile Apps Design

ADAPTERVIEWS AND ADAPTERS

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SELECTIVE WIDGETS

- Lists of views:
  - Static list: ListView,
  - Unfolding list: Spinner,
  - Expandable list: ExpandableListView,
  - Grid: GridView.
- All of them are subclasses of **AdapterView**,
- AdapterView is a subclass of **ViewGroup**

ADAPTERVIEW

- The views (from a list) to be displayed by an **AdapterView** are obtained from an **Adapter**, the interface to a given data structure, in contrast to handling static lists of views.
- The purpose of **AdapterView** is two-fold:
  - To fill out the corresponding layout,
  - To handle a user selection from the list.
- The purpose of **Adapter** is two-fold:
  - Provide the access to the data structure,
  - Provide the individual view of any element of the list.

ADAPTER

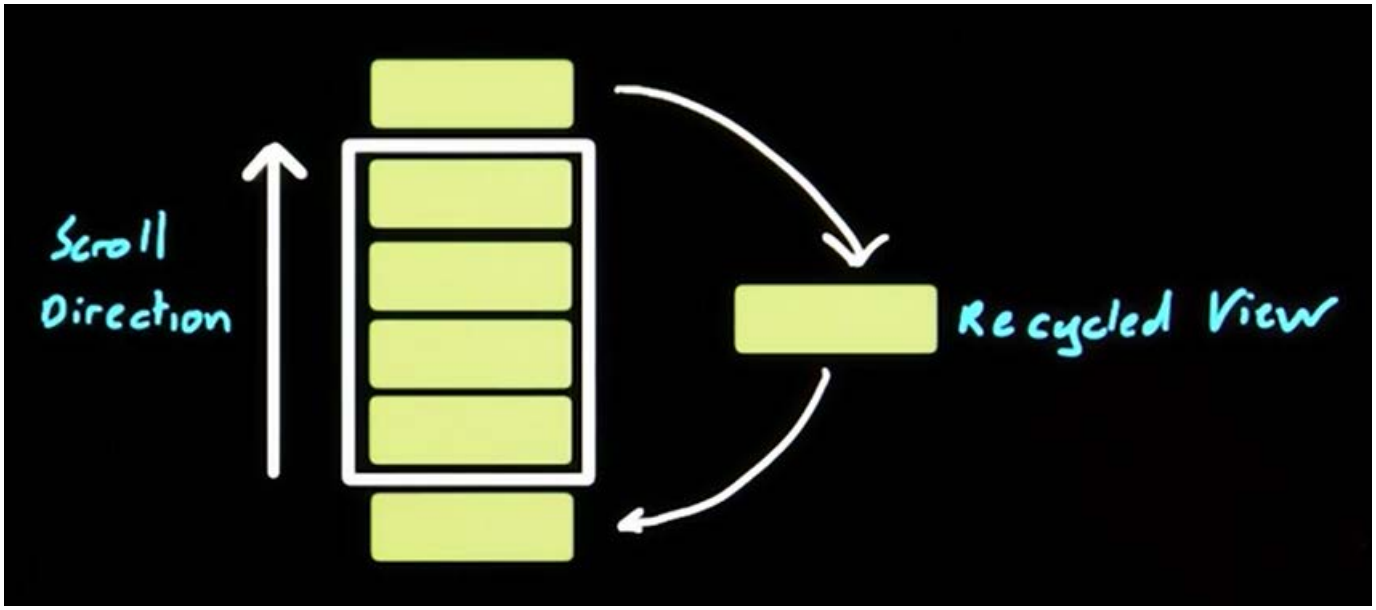
- Implementations of **Adapter** (interface):
  - **BaseAdapter** (abstract class)
  - **ArrayAdapter<T>** (array of arbitrary objects)
  - **SimpleAdapter** (to map static data to views defined in an XML file)
  - **SimpleCursorAdapter** (adapter that exposes data from a Cursor)
  - **SimpleExpandableListAdapter** (map static data to group and child views defined in an XML file)

BASEADAPTER

- To create customized adapters we will extend our Adapter from **BaseAdapter** and overwrite the methods:
  - public int getCount();
  - public View getView(int position, View convertView, ViewGroup parent);
  - public Object getItem(int arg0);
  - public long getItemId(int arg0);

BASEADAPTER

- The **convertViews** are constantly recycled in order to use only as many views as visible on the screen.



EXERCISE

- Complete the code of **MyAdapter_users**
- Complete the code of **d_UsersListActivity** to check the proper programming of MyAdapter_users as well.

EXERCISE

- Complete the code of **MyAdapter_messages** accordingly.
- Complete the code of **e_MessagesActivity** to check the proper programming of MyAdapter_messages as well.